Digital Systems

Lecture 13

Device Drivers

Mark van der Wilk

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UNIVERSITY OF OXFORD

Department of COMPUTER SCIENCE

Following the course of past years by Mike Spivey

List of Demands

We have a single CPU that can execute instructions.

We want to be able to

- execute multiple programs simultaneously (concurrency),
- that can respond to external events and communicate with one another,
- with each program written using its own control flow,
- seeming as each program is running simultaneously,
- **and doesn't require the programmer to know when to disable interrupts in order to guarantee correct code.**

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- Other processes communicate with hardware through device driver, by sending messages.
- Device driver is a process that serves messages from hardware/software in a loop.

Serial Output

```
void serial_putc(char ch) {
  message m;
  m.int1 = ch;
  send(SERIAL, PUTC, &m);
}
```

- Now interacts with hardware via the driver, rather than directly.
- Request message sent to SERIAL driver.
- Process yields to the OS, and waits if driver is not yet ready.

Driver Process

```
void serial_task(int arg) {
  static char txbuf[NBUF];  // Note: State is local to driver!
  int bufin, bufout, bufcount; message m; char ch;
  serial_setup();
  while (1) {                // Loop serves request
    receive(ANY, &m);
    switch (m.m_type) {
      ...
    }
  }
}
```

Setting Things Up

```
void serial_setup(void) {
  ...
  connect(UART_IRQ);
  enable_irq(UART_IRQ);
  UART_INTENSET = BIT(UART_INT_TXDRDY);
}
```

Handling PUTC Messages

```
while (1) {
  receive(ANY, &m);
  switch (m.m_type) {
    case PUTC:
      ch = m.int1;
      txbuf[bufin] = ch; ... // rest of circular buffer
      break;
    ...
  }
}
```

- Buffer variables are local, so no other process can interfere.
- No need for `volatile` or to disable interrupts.

Questions

Several questions remain to be answered:

1. How do characters get out of the buffer and into the UART?
2. What happens about interrupts?
3. What happens when someone wants to send a PUTC message but the buffer is full?

Handling Interrupts

Remember: Hardware sends INTERRUPT message.

Inside loop, inside switch…

```
case INTERRUPT:
  if (UART_TXDRDY) {
    txidle = 1;
    UART_TXDRDY = 0;
  }
  clear_pending(UART_IRQ);
  enable_irq(UART_IRQ);
  break;
```

Responding to Events

Send character, if possible, after any message.

```
while (1) {
  receive(ANY, &m);
  switch (m.m_type) {
    ...
  }
  if (txidle && bufcount > 0) {
    UART_TXD = txbuf[bufout]; ...
    txidle = 0;
  }
}
```

Use message mechanism to synchronise processes!

Full Driver

```
while (1) {
  if (bufcount == NBUF)
    receive(INTERRUPT, &m);
  else
    receive(ANY, &m);
  switch (m.type) {
  case INTERRUPT:
    if (UART_TXDRDY) {
      txidle=1;UART_TXDRDY=0;}
    clear_pending(UART_IRQ);
    enable_irq(UART_IRQ);
    break;
  case PUTC:
    txbuf[bufin] = m.int1;
    bufin = wrap(bufin+1);
    bufcount++;
    break;
  }
  if (txidle && bufcount>0) {
    UART_TXD = txbuf[bufout];
    bufout = wrap(bufout+1);
    bufcount--;
    txidle = 0;
} } }
```

Omitted Here

Lab 4 has a more elaborate serial driver

- Supports both output and input with echoing and line editing.
- All UART initialisation details are filled in.
- There's an alternative interface print_buf that overcomes the one-message-per-character bottleneck.

Bottleneck

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-
-
-
-

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- Allow `printf()` to accumulate characters in a buffer, and send them all at once.
- But… shared memory!

Standard Interrupt Handler

```c
void default_handler(void) {
  int irq = active_irq();
  int task = os_handler[irq];
  disable_irq(irq);
  interrupt(task);    // Send an interrupt message to process
}
```

- Driver can be ready to accept message
  → context switch
- Driver can not be ready to accept message
  → deliver interrupt message later, at next available opportunity

Why do we need `disable_irq(...)`?

Example

Heart & Primes example. Can we find sequence of events? When we switch contexts, when CPU sleeps/wakes?

- Look at `ex-heart.c`.
- Look at `serial.c`

If time, look at `print_buf()`.

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? How does the OS switch tasks?

- How does it decide what to run next?
- How does it move messages?
- How does it actually do the context switch?